

ITEC 4366 Advanced Web Development

Lab 1: Modern Web Development Warmup

Purpose	Refresh core HTML, CSS, responsive design, accessibility, local development workflow, and developer accountability before moving into PHP, data integrations, and server side work.
Estimated Effort	3 to 5 hours, depending on prior experience and design complexity.
Submission Format	One compressed .zip file submitted to the D2L assignment dropbox.
Folder Name	LastNameLab1, such as StinesLab1.
Point Value	25 points

Background

This first lab is a warmup. The goal is not to build the most complicated website in the class. The goal is to make sure your development environment works, your HTML and CSS habits are solid, and your first design gives us something realistic to improve in later labs.

You will build a one page website that can grow into a small multi page site in Lab 2. The topic may be real, fictional, professional, creative, or delightfully weird, as long as it gives you enough content to meet the requirements below.

Learning Goals

- Set up and test a local development environment for PHP and MySQL work later in the course.
- Create a valid HTML5 page using semantic structure and organized content.
- Use modern CSS practices, including responsive layout, reusable classes, and mobile friendly design.
- Apply a front end framework or custom CSS thoughtfully rather than dropping in styles without purpose.
- Test the page in the browser and validate the HTML before submission.
- Use AI tools responsibly, when helpful, while maintaining full developer ownership of the submitted work.

Required Tools

- A code editor such as Visual Studio Code, Notepad++, or another editor you are comfortable using.
- A current web browser such as Chrome, Edge, Firefox, or Safari.
- XAMPP is recommended for local Apache and MySQL support. A Linux VM, Docker based environment, Windows IIS, or another local web server is acceptable if you can demonstrate that it works.
- The W3C HTML Validator for checking your rendered page source.
- Optional but encouraged: browser developer tools, Lighthouse, GitHub, and GitHub Desktop or Git command line tools.

Task 1: Prepare Your Local Development Environment

1. Follow the Installing XAMPP video or use an equivalent local development setup approved by your own technical judgment.
2. Start Apache and MySQL. In XAMPP, both services should show as running and display their port numbers.

3. Inside the web root folder, create a folder named LastNameLab1. For XAMPP, this will usually be inside /xampp/htdocs.
4. Take a screenshot showing your local server environment running. In XAMPP, this should show Apache and MySQL running, with their port numbers visible.
5. Save the screenshot inside your LastNameLab1 folder using a clear filename such as server-running.png.

Note: You are not required to use XAMPP if you already have another working local development environment. However, your screenshot must clearly show that your local server setup is working.

Task 2: Build a One Page Website

Create a homepage that introduces your selected topic and gives the site enough structure to become a small multi page project in future labs.

- Create an index.php file and a style.css file inside your LastNameLab1 folder.
- Open the page through your local web server, not by double clicking the file. Example: `http://localhost/LastNameLab1/`
- Use original structure, writing, images, and design choices that fit your selected topic.

Required Page Content

- Use the HTML5 doctype.
- Include a viewport meta tag for responsive design.
- Link to your external style.css file.
- Use semantic HTML elements such as header, nav, main, section, aside, article, and footer where appropriate.
- Include at least three heading elements using at least two heading levels.
- Include at least three paragraph elements with meaningful content, not placeholder text only.
- Include at least two lists. These may be unordered, ordered, or description lists.
- Include at least two images. Use appropriate file names and alt text.
- Include at least one table that uses thead and tbody sections.
- Include navigation links for possible future pages that could be added in Lab 2.

Required Design and Styling

- Use Bootstrap 5, W3.CSS, or custom CSS. You may combine a framework with your own CSS, but your own style.css file must contain meaningful custom styling.
- Apply a clear color scheme that fits the topic and remains readable.
- Style the navigation so that it functions like a real site menu.
- Create a layout that uses at least two columns on larger screens. CSS Grid, Flexbox, Bootstrap grid classes, or W3.CSS responsive classes are acceptable.
- Make the layout responsive so it works on smaller screens. A two column desktop layout should become comfortable to read on mobile.
- Style the table so it is readable and responsive. The header should stand out, and the body rows should use zebra striping or another clear visual pattern.
- Use spacing, alignment, and visual hierarchy intentionally. The page should look designed, not merely assembled.

Advanced Web Development Habits to Practice

Even though this is a warmup lab, start practicing the habits expected in advanced web development work:

- Accessibility: images should have helpful alt text, links should be descriptive, and color choices should maintain reasonable contrast.
- Responsive thinking: test the design at desktop and mobile widths using browser developer tools.
- Maintainability: keep HTML structure clean, use class names that make sense, and avoid excessive inline styles.
- Performance awareness: use reasonably sized images and avoid adding huge libraries or assets you do not need.
- Developer accountability: test your work and be prepared to explain what your code does.

Validation and Testing

1. View your page through localhost and confirm that the page loads correctly.
2. Right click the rendered page in the browser and choose View Page Source.
3. Copy the rendered HTML source into the W3C Validator direct input option.
4. Fix HTML validation errors that are within your control.
5. Test the page at a narrow mobile width using browser developer tools.
6. Take a screenshot of the website running at localhost. The screenshot must show the browser URL and your website.

The W3C HTML Validator may not accept .php files directly. Viewing the rendered source and pasting it into Direct Input is the preferred approach.

Use of AI Tools

AI tools are allowed in this lab. You may use them to brainstorm topic ideas, explain errors, suggest layout approaches, help debug code, or review your work. However, you are ultimately responsible for the accuracy, functionality, security, originality, and completeness of everything you submit.

Submitting AI generated code or content without understanding, testing, adapting, and documenting it is not acceptable. You should be able to explain your design choices, page structure, CSS rules, and how your site works. AI is a development aid, not a replacement for your own judgment as the developer.

If you use AI in a meaningful way, include a short AI use note in your submission folder. This may be a simple text file named ai-use-note.txt with 3 to 5 sentences explaining what you used AI for and how you reviewed or changed the output before submitting your work.

What to Submit

Submit one .zip file containing your complete LastNameLab1 folder. Your folder should include:

- index.php
- style.css
- all image files used by your page
- a screenshot showing your local server environment running

- a screenshot showing your website running through localhost with the URL visible
- optional but encouraged: ai-use-note.txt if AI tools were used in a meaningful way
- optional but encouraged: a short readme.txt describing your topic and any known issues

Instructor Notes

- Tutorial videos are guides, not the answer key. Your final page should reflect your own design choices.
- I will take off for HTML errors that should reasonably be fixed at this level.
- CSS validator results are useful for debugging, but framework specific CSS can sometimes create warnings that are less important than whether your own CSS is organized and functional.
- The strongest submissions will look like a real starting point for a small website, not a checklist pasted onto a blank page.